

PRISM: Benchmarks for HPCs utilizing Convolutional Neural Networks Model Training

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Abstract

Benchmarking plays a vital role in understanding the performance of supercomputing systems, especially when deciding how to allocate resources to a certain project or job [1]. The role of a benchmark is to measure the performance of a script under different conditions, and by analyzing benchmarks, researchers can determine the optimal environment for the script.

This research project creates a multi-framework system for benchmarking Convolutional Neural Network (CNN) model tasks on supercomputers, including support for both PyTorch and TensorFlow. These benchmarks create a valuable baseline in perceiving how the efficiencies of different libraries change as the codebase evolves, as well as being used as a measure of competency between numerous different supercomputers.

Introduction

PRISM, or Performance Rater Intended for Supercomputer training Models, is a multi-framework dynamic benchmarking system meant for gauging the progress of different frameworks and supercomputers, as well as comparing them through specific tasks. Due to the unique specifications, models, and even architectures of every supercomputer, gauging the efficacy of different computers is near-impossible without a benchmark. PRISM's multi-framework system allows for an expanded range of use, making it more robust as well as allowing the user to determine the best framework for the task at hand.

Materials and Methods

The Expanse Supercomputer was utilized to host PRISM, using the supercomputer's SLURM to schedule and run each of the benchmarks. The two CNN models supported by these benchmarks are PyTorch & TensorFlow, and have been built upon customization, allowing a user to specify model aspects through argparse like precision, epochs, batch size, image size, etc. The models also support saving to ONNX format, the standard format for both training & inference.

PRISM is automated through a central script that runs each model with various allocated CPU amounts, recording the benchmark times to later perform analysis on. The script employs the subprocess package to execute shell scripts, calling SLURM job arrays to account for variation in individual benchmark runtimes.

PRISM also uses a custom-made dataset (SDSC-10) based off the ImageNet dataset. The sizes are standardized at 192 by 128, and cropped using a rolling window and edge density algorithm. Over 300,000 images are contained within the SDSC-10 dataset.

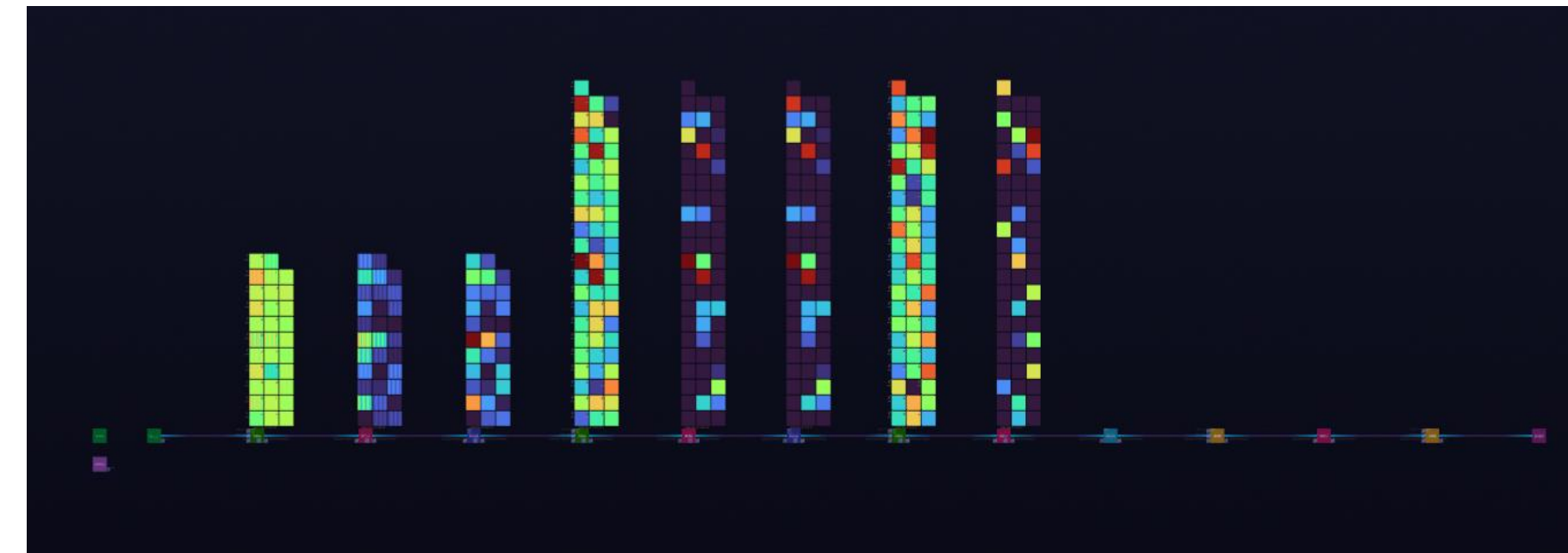


Figure 1: Visualization of a model generated by PRISM during benchmarking

Results

The results of PRISM, illustrated in Figures 2-7, show how both the speedup and the duration tapered out as cores increased. For both PyTorch and TensorFlow, the increase in cores maximized its speedup, and in turn minimized the duration, but the speedup rate greatly diminished from 32-64 cores. The speedup was more efficient for TensorFlow, while duration was similar for both. Furthermore, the GPU benchmark was significantly faster than all the different CPU benchmarks for both TensorFlow and PyTorch.

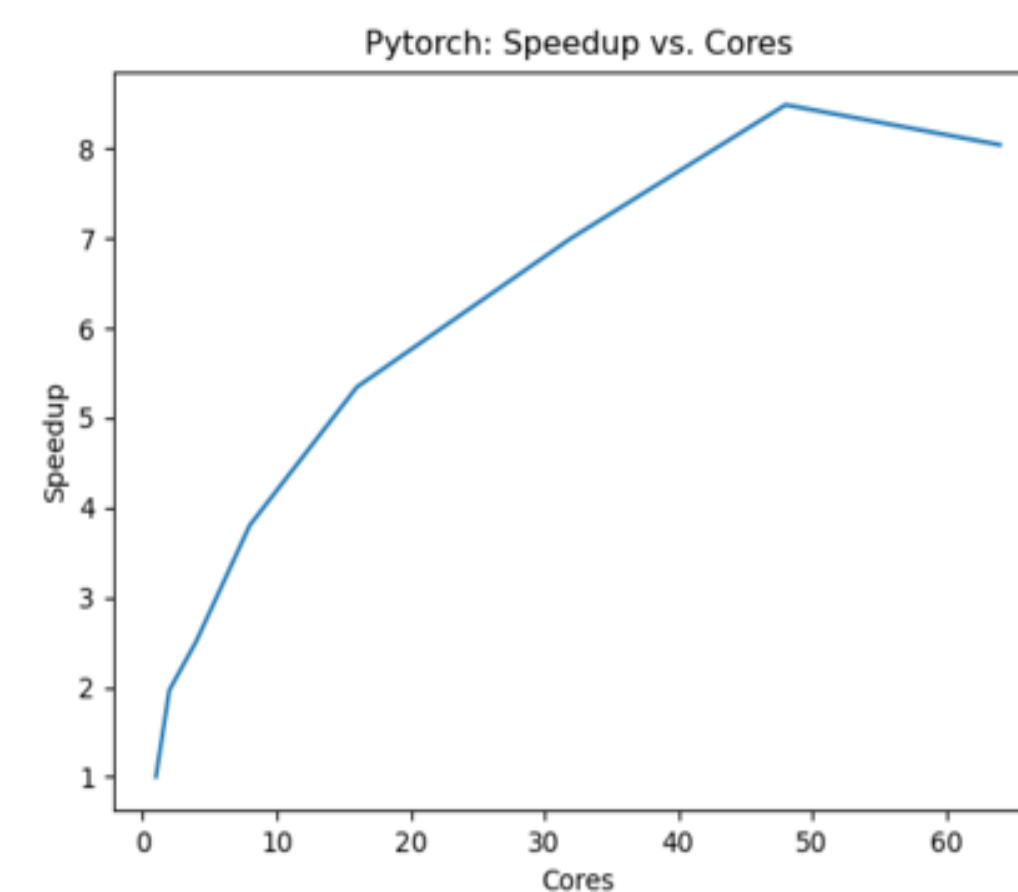


Figure 2

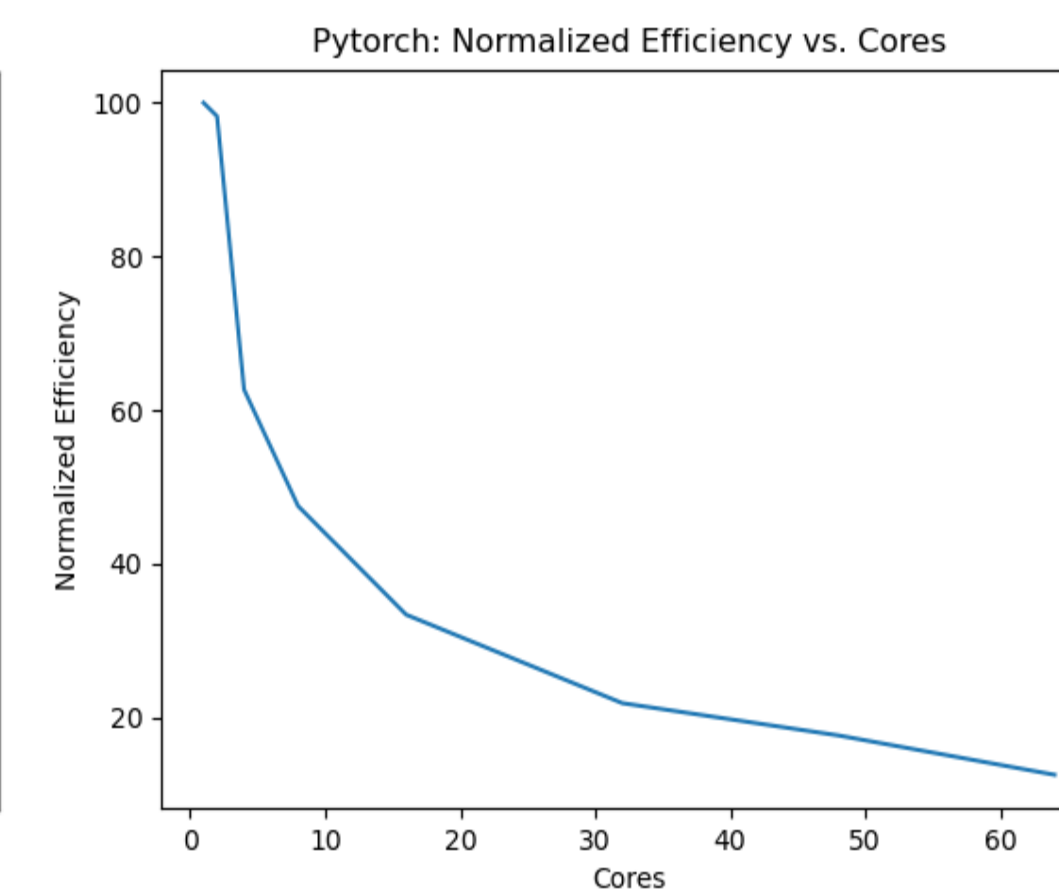


Figure 3

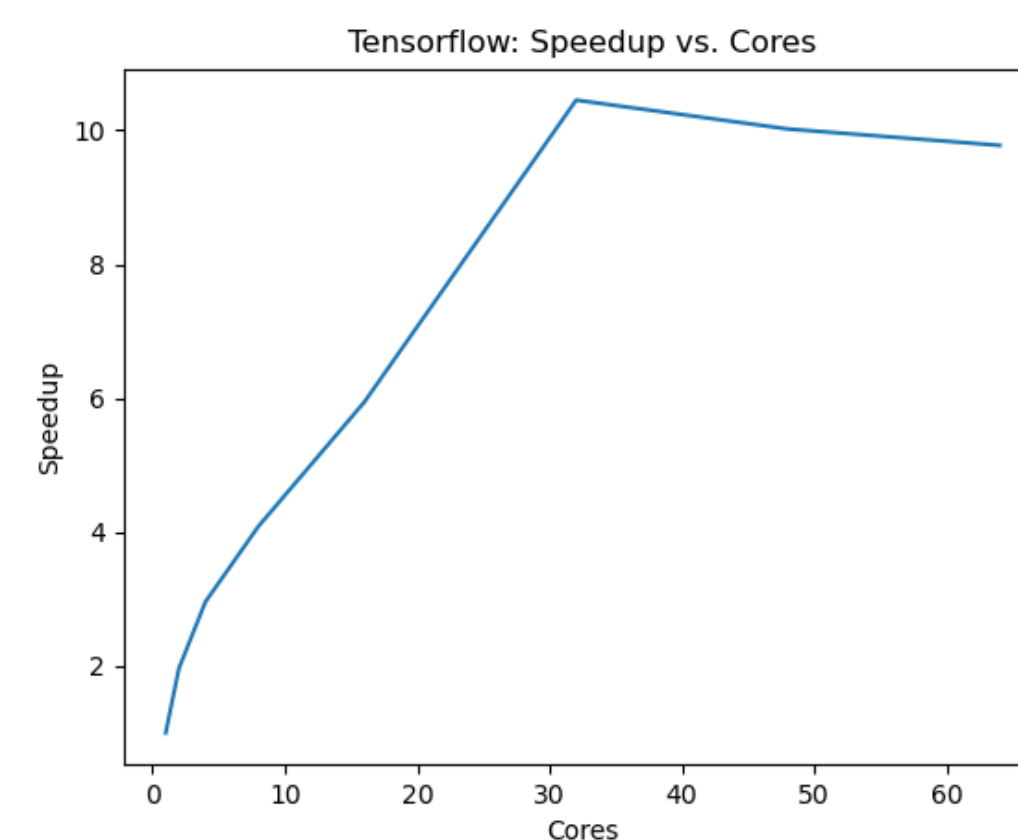


Figure 4

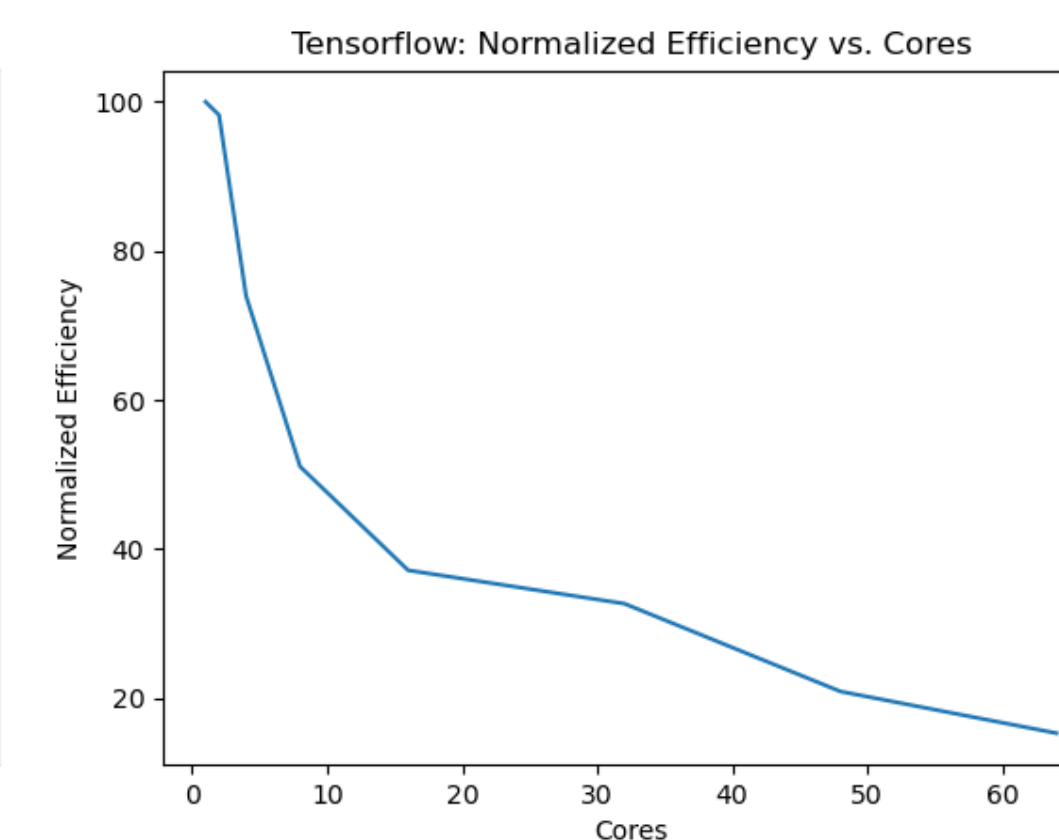


Figure 5

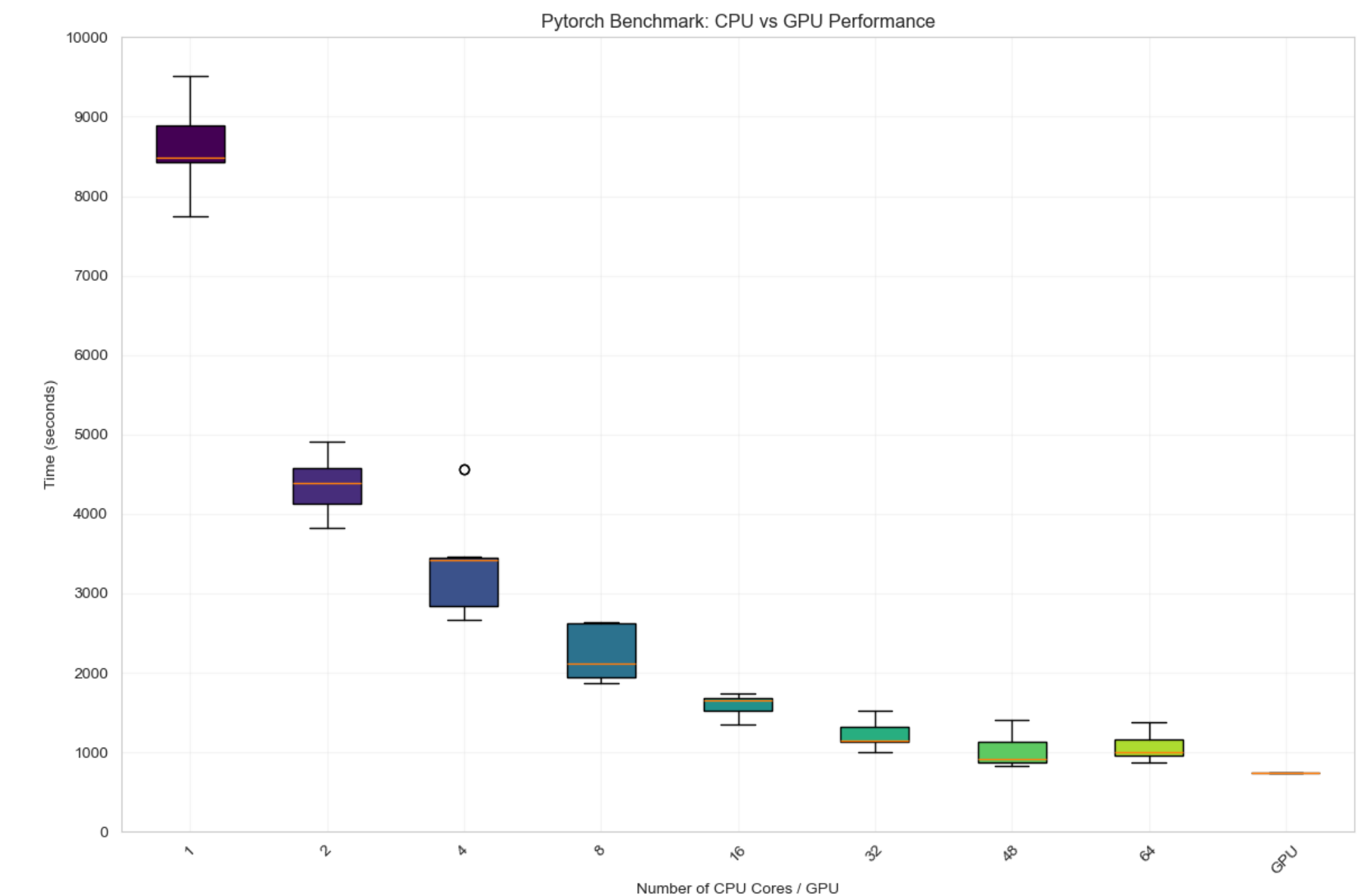


Figure 6

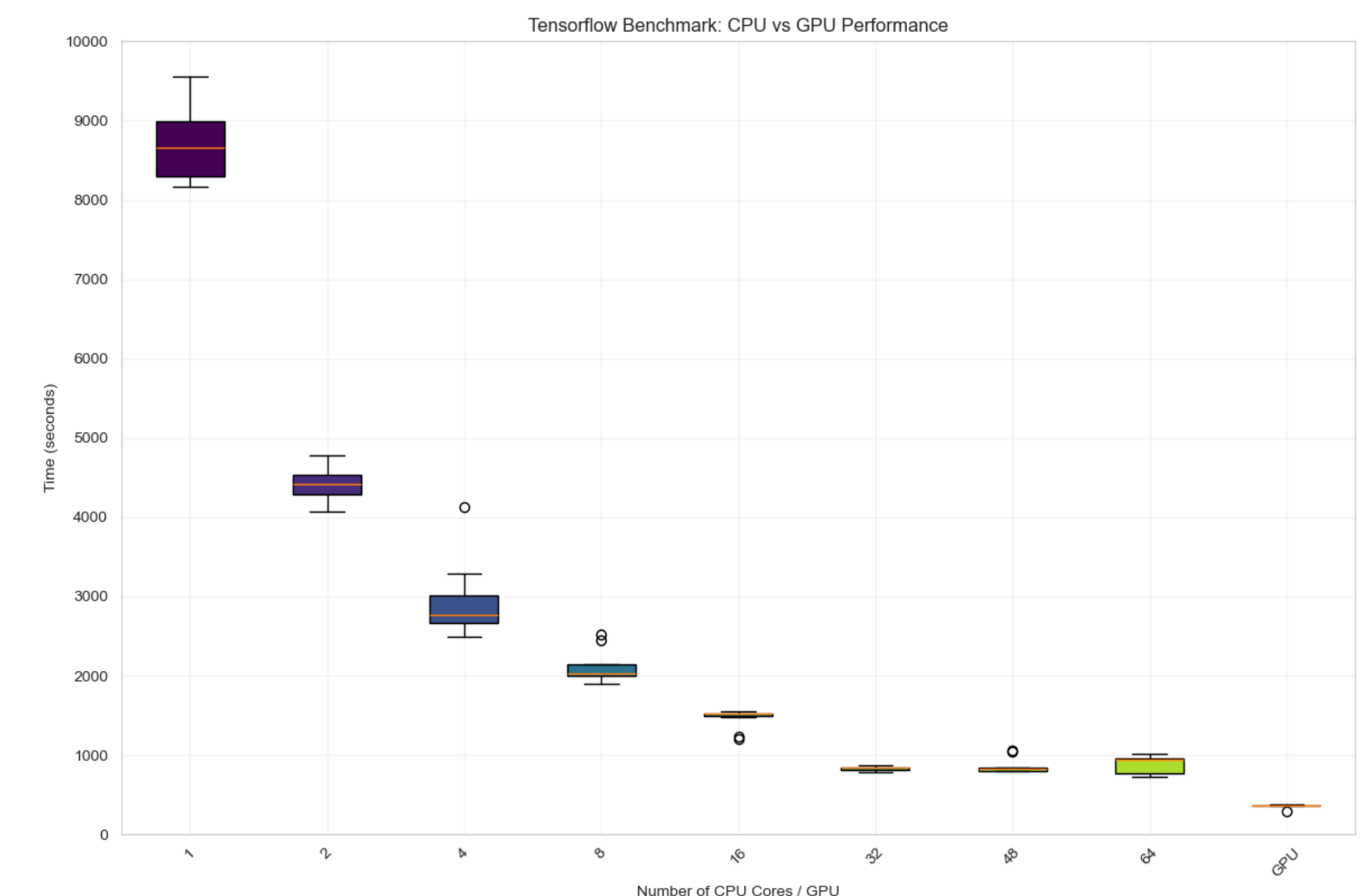


Figure 7

Applications & Future Work

PRISM can be used to gauge the efficacy of certain optimizations within PyTorch & TensorFlow scripts. This can create a standardized way to accurately gather data, making decisions easier for researchers. PRISM can be further expanded upon through the implementation of more frameworks, benchmarking, and datapoints.

References

[1] Ciżnicki, M., Kierzyńska, M., Kopta, P., Kurowski, K., & Gepner, P. (2012). Benchmarking data and compute intensive applications on modern CPU and GPU architectures. *Procedia Computer Science*, 9, 1900–1909. <https://doi.org/10.1016/j.procs.2012.04.208> [2] H. Simon and E. Nehru, "A case for the World Wide Web," in *Proceedings of the Workshop on Data Mining and Knowledge Discovery*, July 2005.